Is here still where it was? Changes in terrestrial reference frames in R spatial packages

useR! 2026, Warsaw, Poland

Roger Bivand

2026-07-08

Norwegian School of Economics, Bergen, Norway

Slides

https://rsbivand.github.io/new_ref_systems

Where is here?

Introduction

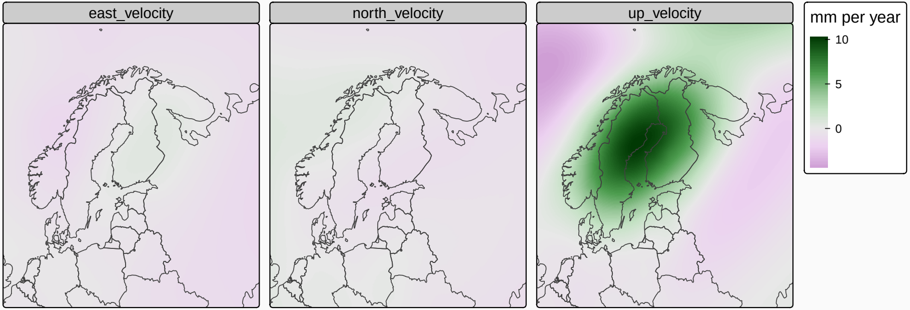
- When we represent or analyse spatial data, the position of observations matters
- Where objects of interest are, also in relation to each other, and how position is measured, are referenced through coordinate reference systems (CRS), including units of measurement
- Frequently used CRS rely on tabulated values, for example the axis lengths of an ellipsoid representing the Earth
- The tabulations crucially include *datums*, including the ellipsoid used in their defintion
- Historical datums (typically tied to mean sea level at a well-known location) were established before continental drift was accepted (roughly 1960s)

Continental drift

- The measurement of continental drift – plate tectonics – became feasible, and even obvious, when enough satellites could be used
- Geodetic measurements in metres from the 3D centre of the Earth are now the most reliable and accurate basis for recording object position
- "Stationary" objects move in geodetic terms when the tectonic plate on which they are located moves in any of the three axes
- Plate movements are anchored by date, so position measurements need to be dated to fix them in 3D+time
- "Here" in the view of one country's mapping agency may not have the same coordinates as the same position viewed by a neighbouring country's mapping agency, or the same country's mapping agency at a different point in time

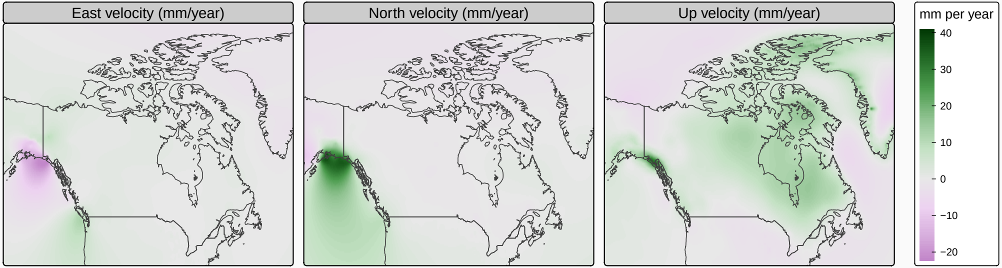
Nordic-Baltic velocity grids

A velocity grid by the Nordic Geodetic Commission provides a deformation model covering the Nordic and Baltic countries, used in transformations between global reference frames and the local realizations of ETRS89 (European Terrestrial Reference System 1989) in the Nordic and Baltic countries (https://cdn.proj.org/eur_nkg_README.txt, Häkli et al. (2023)), read with `stars` (Pebesma 2026b) and displayed using `tmap` (Tennekes 2026)



Canadian velocity grid

`N2022v00VG.gvb` is a beta release velocity grid, created prior to forthcoming changes in North America known as NATRF2022, downloaded from (login protected) the Canadian Geodetic Survey



Preliminary take-home points

- Most legacy CRS did not take continental drift into account
- The demands on CRS for accuracy and measurable errors have increased greatly, especially with the general availability of GNSS receivers
- Countries/jurisdictions adopt CRS for official purposes, and often manage them differently
- Global CRS have to match global crustal movement, but regional/national CRS are most often fixed to the most appropriate tectonic plate
- Errors at metre scale are now unacceptable, and sub-metre errors are undesirable

Definitions

SDSR definitions i

In Pebesma and Bivand (2023), page 22-3, “we follow(ed) Lott (2015) when defining the following concepts (italics indicate literal quoting, here updated to Lott (2023a)):

- a **coordinate system** (§4.1.8) is a *set of mathematical rules for specifying how coordinates are to be assigned to points*,
- a **datum** (§4.1.11) is a *parameter or set of parameters that define the position of the origin, the scale, and the orientation of a coordinate system*, (in Lott (2023a) (§3.1.12), referred to as a reference frame, a realization of a reference system)
- a **geodetic datum** (§4.1.19) is a *datum describing the relationship of a two- or three-dimensional coordinate system to the Earth*, and

- a **coordinate reference system** (§4.1.7, **CRS**) is a *coordinate system that is related to an object by a datum; for geodetic and vertical datums, the object will be the Earth*; Lott (2023a) (§3.1.8) adds a note that geodetic and vertical datums are referred to as reference frames.

A readable text that further explains these concepts is(was) Iliffe and Lott (2008)."

What is an *epoch* and a *datum ensemble*? i

- Iliffe and Lott (2008) do not discuss *epoch* or *datum ensemble*, which entered the European Petroleum Survey Group (EPSG) database at version v10 (September 2020) and subsequently, nor does Lott (2015), but Lott (2019) and Lott (2023a) do; Iliffe and Lott (2008) and Lott (2015) follow ISO 19111:2007 and ISO 19111-2:2009 which apply mainly to *static* CRS (dynamic is not mentioned in Lott (2015)), while Lott (2019) and Lott (2023a) elevate to 19111:2019 which also handles *dynamic* CRS (dynamic is mentioned 125 times in Lott (2023a))
- Lott (2023a) (§3.1.22) defines **epoch** as a "*<geodesy> point in time, an epoch is expressed in the Gregorian calendar as a decimal year*"; it expresses the moment at which a position in 2D or 3D is valid with respect to the realization of the reference system, geodetic datum

What is an *epoch* and a *datum ensemble*? ii

- (§7.7) "A CRS is *dynamic* if its reference frame is dynamic. In a *dynamic* CRS the coordinate values of a point on or near the surface of the Earth change with time. When a coordinate set is referenced to a dynamic CRS, to be unambiguous the coordinate set needs to additionally be referenced to a coordinate *epoch*.

What is an *epoch* and a *datum ensemble*? iii

- A **datum ensemble** (§3.1.13) is a "*group of multiple realizations of the same terrestrial or vertical reference system that, for approximate spatial referencing purposes, are not significantly different*"; Lott (2023a) notes that using WGS84 as example, "*an undifferentiated group of realizations including WGS 84 (TRANSIT), WGS 84 (G730), WGS 84 (G873), WGS 84 (G1150), WGS 84 (G1674) and WGS 84 (G1762). At the surface of the Earth these have changed on average by 0.7 m between the TRANSIT and G730 realizations, a further 0.2 m between G730 and G873, 0.06 m between G873 and G1150, 0.2 m between G1150 and G1674 and 0.02 m between G1674 and G1762)*. It further notes that:" *'Approximate' is for users to define but typically is in the order of under 1 decimetre but may be up to 2 metres*". G-numbers are GPS weeks.

WGS84 datum ensemble i

WGS84, also known as EPSG:4326, is perhaps the most frequently used CRS, but is actually an ensemble as noted above and retreived from `sf::st_crs("EPSG:4326")` (Pebesma 2026a):

```
##     ENSEMBLE["World Geodetic System 1984 ensemble",
##         MEMBER["World Geodetic System 1984 (Transit)"],
##         MEMBER["World Geodetic System 1984 (G730)"],
##         MEMBER["World Geodetic System 1984 (G873)"],
##         MEMBER["World Geodetic System 1984 (G1150)"],
##         MEMBER["World Geodetic System 1984 (G1674)"],
##         MEMBER["World Geodetic System 1984 (G1762)"],
##         MEMBER["World Geodetic System 1984 (G2139)"],
##         MEMBER["World Geodetic System 1984 (G2296)"],
##         ELLIPSOID["WGS 84",6378137,298.257223563,
##             LENGTHUNIT["metre",1]],
```

WGS84 datum ensemble ii

```
##          ENSEMBLEACCURACY[2.0]],
```

Table 1: WGS84 iterations

Name	GEOD	GEOG	Epoch	Based.on	Implemented
WGS84(TRANSIT)	7815	7816	NA	BTS84	1987-01-01
WGS84(G730)	7656	7657	1994	ITRF92	1994-06-29
WGS84(G873)	7658	7659	1997	ITRF94	1997-01-29
WGS84(G1150)	7660	7661	2001	ITRF2000	2002-01-20
WGS84(G1674)	7662	7663	2005	ITRF2008/IGS08	2012-02-08
WGS84(G1762)	7664	7665	2005	ITRF2008/IGb08	2013-10-16
WGS84(G2139)	9753	9754	2016	ITRF2014/IGS14	2021-01-03
WGS84(G2296)	10604	10605	2024	ITRF2020/IGS20	2024-01-07

WGS84 datum ensemble iii

See also ESRIs blog “WGS 1984 is not what you think!”, and Konyk et al. (2025)

The most recent update is aligned with ITRF2020, the most recent International Terrestrial Reference Frame (ITRF) realization; it is now defined as dynamic

```
## GEODCRS["WGS 84 (G2296)",
##     DYNAMIC[
##         FRAMEEPOCH[2024]],
##     DATUM["World Geodetic System 1984 (G2296)",
##         ELLIPSOID["WGS 84",6378137,298.257223563,
##             LENGTHUNIT["metre",1]]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[Cartesian,3],
##         AXIS["(X)",geocentricX,
##             ORDER[1],
```

WGS84 datum ensemble iv

```
##                 LENGTHUNIT["metre",1]],
##             AXIS["(Y)",geocentricY,
##                 ORDER[2],
##                 LENGTHUNIT["metre",1]],
##             AXIS["(Z)",geocentricZ,
##                 ORDER[3],
##                 LENGTHUNIT["metre",1]],
##         USAGE[
##             SCOPE["Geodesy. Navigation and positioning using GPS satellite system."],
##             AREA["World."],
##             BBOX[-90,-180,90,180]],
##         ID["EPSG",10604],
##         REMARK["Replaces WGS 84 (G2139) (CRS code 9753) from 2024-01-07."]]
```

From historical datums to reference frame realizations

- An International Terrestrial Reference Frame (IRTF) is a realization of the International Terrestrial Reference System (ITRS), which describes procedures for creating reference frames suitable for use with measurements on or near the Earth's surface
- Each realization attempts to improve on earlier iterations, as mathematical, computational and surveying techniques improve, and as measurements of the speed of shifts in the Earth's surface in relation to the geoid and ellipsoid surface advance
- More accurate measurements of gravitation and magnetism contribute to this, through both satellite and terrestrial instrumentation
- Many grids are now available in the PROJ CDN to transform between epochs, based on other grids showing velocities of change between global reference frames and local realizations

Software implementation in R spatial packages

EPSG, GDAL (>=3) and PROJ (>=6)

- Most R geospatial packages (`sf`, `terra` (Hijmans et al. 2026), `gdalraster` (Toney and Sumner 2026), `vapour` (Sumner 2026), `gdalcubes` (Appel 2026)) use the PROJ external library (Evenden et al. 2026), its database `proj.db` and transformation grid files from the PROJ CDN (content download network) through the GDAL external library (Rouault et al. 2026); `geographiclib` (Sumner and Karney 2026) does not use `proj.db`

- This relationship between geospatial packages and `proj.db` has existed since GDAL (>=3) and PROJ (>=6) in 2019, and the PROJ CDN (https://cdn.proj.org from PROJ (>=7) in 2020)

- The `proj.db` SQLite database contains tables from the EPSG (https://iogp.org, https://epsg.org) geodetic parameter dataset

- Both the EPSG geodetic parameter dataset and the database layout used by PROJ to construct `proj.db` are versioned, raising questions about handling

Recent `proj.db` version information i

Treating `proj.db` as an SQLite database, we can extract information from the `metadata` table. Version v12.029 is provided in PROJ 9.7.1 (December 2025) and 9.8.1 (April 2026, https://proj.org/en/stable/news.html), v12.049 was tried in 9.8.0 (March 2026) but caused trouble - more on this below, and the development version of PROJ (commit d258a97, 19 June) due for release in September 2026 provides currently v12.057:

Table 2: Recent PROJ `proj.db` characteristics

PROJ	epsg	layout_minor	tables
9.7.1	v12.029	6	47
9.8.0	v12.049	6	47
9.9.0	v12.057	7	48

Recent `proj.db` version information ii

The difference between the two newer databases is the addition of a new table: derived_projected_crs, causing an increment in the layout minor version. PROJ fails if the layout version of `proj.db` and that expected by the external library differ.

`proj.db` is updated at releases of PROJ, and made available both through package managers for Linux source installs, and through updates to R build trains for macOS and Windows (Rtools).

New standards are entering the EPSG database, so through PROJ and GDAL, enter R workflows, and as this happens, the same intended coordinate transformation may yield different results, depending on CRS definitions, EPSG versions and *epochs*

Recent `proj.db` version information iii

The adaptations to earlier changes in the PROJ external library described in section 4 of Bivand (2021) and Pebesma and Bivand (2023) may not be sufficient, see also https://r-spatial.github.io/evolution/CRS_projections_transformations.html, which is also no longer sufficient

EPSG table changes disrupt workflows

Prelude in Finland: `geofi` started failing with PROJ 9.6.0 i

- Before the release of PROJ 9.6.0 in March 2025, I was surprised to see an error in the R package `geofi` (Kainu et al. 2026) on running reverse dependency checks; `geofi` imports from `sf`
- `geofi` provides some saved geometries, including their CRS, and something had gone wrong with the intersection of a saved geometry and one freshly downloaded with a CRS instantiated from PROJ 9.6.0 - previously the CRS of the saved and freshly instantiated CRS were identical, now they differed
- At present, macOS binaries of `sf` are static built with PROJ 9.5.1, Windows with PROJ 9.7.1, COPR for RHEL/Fedora-rawhide PROJ 9.8.1, etc.

Prelude in Finland: `geofi` started failing with PROJ 9.6.0 ii

- The issue was resolved (https://github.com/rOpenGov/geofi/issues/52), but the problem was caused by the National Land Survey of Finland changing the definition of the same EPSG 3067 code
- Pasi Häkli explained why they chose this alternative in a useful exchange on the PROJ mailing list; the underlying reason was connected to the emerging need to specify a datum ensemble member precisely (https://lists.osgeo.org/pipermail/proj/2025-March/011787.htm)
- Fortunately, there was a test in `geofi`, so the consequences of a change in `proj.db` stemming from an improvement made by a national mapping agency in a CRS definition were detected

ETRS: ETRF iterations

Name	GEOD	GEOG3D	GEOG2D	Epoch	Based.on
ETRF89	7914	7915	9059	1989,0	ITRF89
ETRF90	7916	7917	9060	1989,0	ITRF90
ETRF91	7918	7919	9061	1989,0	ITRF91
ETRF92	7920	7921	9062	1989,0	ITRF92
ETRF93	7922	7923	9063	1989,0	ITRF93
ETRF94	7924	7925	9064	1989,0	ITRF94
ETRF96	7926	7927	9065	1989,0	ITRF96
ETRF97	7928	7929	9066	1989,0	ITRF97
ETRF2000	7930	7931	9067	1989,0	ITRF2000
ETRF2005	8397	8399	9068	1989,0	ITRF2005
ETRF2014	8401	8403	9069	1989,0	ITRF2014
ETRF2020	10569	10570	10571	1989,0	ITRF2020

GEOD: EPSG code, geocentric (m); GEOG3D: EPSG code, geographical with ellipsoidal height (m); GEOG2D: EPSG code, geographical

Improved datum definitions disrupt searching for transformation pipelines i

The release of PROJ 9.8.0 on 2 March 2026 with EPSG database v12.049 including an in-place update to EPSG:4937, the European Terrestrial Reference System 1989 ensemble, things got complicated.

Jochem Lesparre on the PROJ mailing list (10 March 2026)- “transformations that used to give correct results (e.g. in PROJ 9.7.1) do no longer give results according to the official transformation” ((https://lists.osgeo.org/pipermail/proj/2026-March/012026.html)

If we take the difference between ETRS89 (EPSG:4937) in v12.029 and v12.049, we see that 45 national/local realizations were added to the list of ensemble members(and one other line was edited):

Improved datum definitions disrupt searching for transformation pipelines ii

```
## in v12.049, not in v12.029
##          MEMBER["ETRS89-DNK"],
##          MEMBER["Estonia 1997"],
##          MEMBER["Albanian Geodetic Reference Frame 2010"],
##          MEMBER["ETRS89-ALB [CORS]"],
##          MEMBER["ETRS89-AUT [2002]"],
##          MEMBER["Belgian Reference Frame 2002"],
##          MEMBER["Belgian Reference Frame 2011"],
##          MEMBER["BH_ETRS89"],
##          MEMBER["Bulgaria Geodetic System 2005"],
##          MEMBER["Croatian Terrestrial Reference System 1996"],
##          MEMBER["AGRS2010 (ETRF2000)"],
##          MEMBER["CROPOS"],
##          MEMBER["ETRF2000 Poland"],
##          MEMBER["ETRS89/DREF91 Realization 2016"],
##          MEMBER["ETRS89/DREF91 Realization 2025"],
##          MEMBER["ETRS89-CZE [2007]"],
```

Improved datum definitions disrupt searching for transformation pipelines iii

```
##          MEMBER["ETRS89-ESP [REGENTE]"],
##          MEMBER["ETRS89-ESP [ERGNSS]"],
##          MEMBER["ETRS89-FRO [2008]"],
##          MEMBER["ETRS89-GRC [HTRS07]"],
##          MEMBER["ETRS89-HUN [ETRF2000]"],
##          MEMBER["ETRS89-IRE [ETRF2000]"],
##          MEMBER["ETRS89-MKD [EUREF-MAK2010]"],
##          MEMBER["ETRS89-NOR [EUREF89]"],
##          MEMBER["ETRS89-PRT [1995]"],
##          MEMBER["ETRS89-ROU [ETRF2000]"],
##          MEMBER["ETRS89-SVK [SKTRF09]"],
##          MEMBER["ETRS89-SVK [SKTRF2022]"],
##          MEMBER["EUREF-FIN"],
##          MEMBER["Istituto Geografico Militaire 1995"],
##          MEMBER["Kosovo Reference System 2001"],
##          MEMBER["Latvian geodetic coordinate system 1992"],
##          MEMBER["Latvian coordinate system 2020"],
```

Improved datum definitions disrupt searching for transformation pipelines iv

```
##         MEMBER["Lithuania 1994 (ETRS89)"],
##         MEMBER["MOLDREF99"],
##         MEMBER["OSNet v2009"],
##         MEMBER["Reseau Geodesique Francais 1993 v1"],
##         MEMBER["Reseau Geodesique Francais 1993 v2"],
##         MEMBER["Reseau Geodesique Francais 1993 v2b"],
##         MEMBER["Rete Dinamica Nazionale 2008"],
##         MEMBER["Serbian Reference Network 1998"],
##         MEMBER["Serbian Spatial Reference System 2000"],
##         MEMBER["Slovenia Geodetic Datum 1996"],
##         MEMBER["SWEREF 99"],
##         MEMBER["Swiss Terrestrial Reference Frame 1995"],
## in v12.029, not in v12.049
##     ID["EPSG",4937]]
```

Between v12.049 and v12.057, changes are more cosmetic:

Improved datum definitions disrupt searching for transformation pipelines v

```
## in v12.049, not in v12.057
##         MEMBER["ETRS89-NOR [EUREF89]"],
##         MEMBER["Istituto Geografico Militaire 1995"],
## in v12.057, not in v12.049
##         MEMBER["EUREF89"],
##         MEMBER["Istituto Geografico Militare 1995"],
##         MEMBER["ETRS89-LUX [ETRF2000]"],
##         MEMBER["Nordic Geodetic Commission ETRF14"],
```

The point raised by Jochem Lesparre applied not just to the changes in place in CRS definitions, but more crucially to the lack of synchronization engendered in transformation between CRS. Adding information to CRS to accommodate our understanding that they should more often be seen as dynamic is probably needed.

Improved datum definitions disrupt searching for transformation pipelines vi

However, the growing complexity of datum definitions taking epoch into account has made the heuristics for choosing transformation pipelines (also from tables in `proj.db`) more challenging.

Legacy PROJ 4 used the (now) datum ensemble WGS84 as a static hub for coordinate transformation between a source and a target CRS, transforming from source datum to WGS84, then transforming again to the target datum reducing accuracy twice.

From PROJ 6, only one transformation pipeline is used, directly from source to target or via varying intermediate steps (Knudsen and Evers 2017; Evers and Knudsen 2017).

Improved datum definitions disrupt searching for transformation pipelines vii

Because the code used for searching for transformation pipelines may be updated between versions of PROJ itself, we need not only to track versions of EPSG, but also versions of PROJ.

The pipelines now need to be found by searching multiple table joins on the EPSG database in `proj.db`, ordering in terms of accuracy and whether transformation grids needed by the preferred pipelines are available, for example for on-demand download from the PROJ CDN.

Improved datum definitions disrupt searching for transformation pipelines viii

Even Rouault commented 12 March: “There are situations where if you use the most complicated SQL requests it supports (and PROJ just limits itself to finding a single intermediate step...) you can end up with several hundreds or more potential transformation pipelines, with runtime of the order of second. While the execution of the pipeline is of the order of the microsecond for a coordinate. So before going to that, it checks if the results of less costly requests look to be good enough” (https://lists.osgeo.org/pipermail/proj/2026-March/012037.html).

Improved datum definitions disrupt searching for transformation pipelines ix

Again on 10 May: "#PROJ challenge: propose a projinfo invocation searching operations between a source and target CRS and switches of your choice leading to the highest number of candidate operations. Record to beat: 717 (including ballpark operations) / 399 (without ballpark)" (https://mastodon.social/@EvenRouault/116547361316876154).

PROJ 9.8.0 and `proj.db` based on EPSG v12.049 was replaced by PROJ 9.8.1 with the same `proj.db` based on EPSG v12.029 published in PROJ 9.7.1.

The development version of PROJ has, since the problems became evident, received numerous "MEGA" (Make ETRS89 Great Again) commits.

Improved datum definitions disrupt searching for transformation pipelines x

The discussion in https://github.com/OSGeo/PROJ/pull/4736 summarizes the situation as "... *I hope the EPSG guys come up with a solid plan. At the moment they're testing in prod(uction)*" (Kristian Evers, 3 April).

PROJ PR https://github.com/OSGeo/PROJ/pull/4773 merged 14 May adopts a hopefully less fragile logic for finding pipeline candidates affected by the ETRS89 ensemble changes in the EPSG database.

Does the remark by Jochem Lesparre (12 March) finding errors of up to 0.25 m affect R workflows when comparing coordinate transformation pipelines chosen in PROJ 9.7.1 and the troubled PROJ 9.8.0; is the problem mitigated in PROJ devel? See https://lists.osgeo.org/pipermail/proj/2026-March/012036.html

Improved datum definitions disrupt searching for transformation pipelines xi

Running the command to identify candidate transformation pipelines as in Jochem Lesparre's post: `sf::sf_proj_pipelines("EPSG:28992", "EPSG:4258", Use="INTERSECTION")` on these three versions of PROJ, we find that 9.7.1 returns 2 candidates, 9.8.0: 4 candidates, and 9.9.0: 9 candidates. So candidate counts increased as heuristics were modified to compensate for the added complexity of the epoch-based ETRS89 datum ensemble.

```
> (m_71_80 <- match(ot_9.7.1_int[,"definition"], ot_9.8.0_int[,"definition"]))
## [1] NA NA
> (m_71_90 <- match(ot_9.7.1_int[,"definition"], ot_9.9.0_int[,"definition"]))
## [1] 1 2
> ot_9.9.0_int[m_71_90, "accuracy"]
## [1] 0.101 0.350
```

Neither of the two pipelines found for 9.7.1 were found for 9.8.0, meaning that the change to the broader ETRS89 ensemble had confused the PROJ $<=$ 9.8.0 heuristics for finding preferred pipelines.

The two 9.7.1 pipelines were found for 9.9.0, where the first gives high accuracy, using a modern grid `rdtrans2018` available from https://cdn.proj.org/nl_nsgi_README.txt, the PROJ CDN:

```
## +proj=pipeline
## +step +inv +proj=sterea +lat_0=52.1561605555556 +lon_0=5.38763888888889 +k=0.9999079
##     +x_0=155000 +y_0=463000 +ellps=bessel
## +step +proj=hgridshift +grids=nl_nsgi_rdtrans2018.tif
## +step +proj=unitconvert +xy_in=rad +xy_out=deg
```

Improved datum definitions disrupt searching for transformation pipelines xiii

The second pipeline is a Helmert transformation. All four 9.8.0 pipelines were found for 9.9.0, but none of them were in the most accurate position (ranked 1).

The first found pipeline for 9.8.0 used an out-dated grid not available from the PROJ CDN, the second a Molobadekas transformation, the third a Helmert transformation, and the fourth only has ballpark accuracy.

```
> (m_80_90 <- match(ot_9.8.0_int[,"definition"], ot_9.9.0_int[,"definition"]))
## [1] 3 4 5 8
```

Using the classical Meuse bank soil pollution data set in the sp package (Pebesma and Bivand 2026), with coordinates measured in EPSG:28992 and tranforming to EPSG:4258 as in Jochem Lesparre's issue, we can track the average distances between transformations made with different versions of PROJ and the associated versions of the EPSG database.

We have 9.7.1 (under Windows, so a slightly different C math library) and 9.8.0 and 9.9.0 d258a97 (Fedora 44). 9.8.0 is about 12 cm off on average by comparison with 9.7.1 at the data location in the far south of the Netherlands. 9.9.0 has the same difference from 9.8.0.

```
> mean(diag(sf::st_distance(t1_9.7.1_cdn, t1_9.8.0_cdn)))
## 0.12438 [m]
> mean(diag(sf::st_distance(t1_9.8.0_cdn, t1_9.9.0_cdn)))
## 0.12438 [m]
```

Using the pipeline argument in `sf::st_transform`, the modern grid pipeline `rdtrans2018` can be replicated, as can the `molobadekas` pipeline. This demonstrates that 9.8.0 uses the `molobadekas` pipeline, 9.7.1 and 9.9.0 use the `rdtrans2018` pipeline, and that 9.7.1 and 9.9.0 differ by a minimal amount between Windows and Fedora C math libraries (the pipelines involve calls to trigonometry functions).

Improved datum definitions disrupt searching for transformation pipelines xvi

```
> mean(diag(sf::st_distance(molobadekas_sf, t1_9.8.0_cdn)))
## 0 [m]
> mean(diag(sf::st_distance(molobadekas_sf, rdtrans2018_sf)))
## 0.12438 [m]
> mean(diag(sf::st_distance(t1_9.9.0_cdn, rdtrans2018_sf)))
## 0 [m]
> mean(diag(sf::st_distance(t1_9.7.1_cdn, t1_9.9.0_cdn)))
## 4.777793e-10 [m]
```

ETRS status

- So, yes, Jochem Lesparre's issue obviously would affect R spatial workflows if packages are built against PROJ 9.8.0
- Luckily, the problem was detected early within PROJ, and PROJ 9.8.1 was released with EPSG v12.029, stepping back from tracking current EPSG moves
- Work is continuing within PROJ to debate, test and implement improvements to heuristics for finding transformation pipelines and ordering then by accuracy
- *Epoch* and *datum ensemble* issues are unlikely to go way, as changes in North America are forthcoming
- Even Rouault on Mastodon (May 13): "Heard in a meeting: proprietary software tracks #PROJ changes to keep informed of changes in the EPSG dataset" https://mastodon.social/@EvenRouault/116567675016927438

Who are the actors?

- Perhaps changes are being driven by national or sub-national mapping agencies; the British Ordnance Survey has a useful guide
- In addition, supra-national coordinating bodies, for example in navigaation, contribute; Eurocontrol has a similar guide
- Cartography and surveying have become much less important as users of position data, while navigation, construction and agriculture are more important civilian uses
- Integration and analysis of attribute data observed at recorded locations over time faces challenges if the CRS of the locations are changed

North America

Forthcoming changes in North America

- Natural Resources Canada reviews the background for Canadian Spatial Reference System (CSRS) modernization
- The North American Datum of 1983 (NAD83) was extended to create NAD83(CSRS) in the late 1990s as a three-dimensional reference system, and with subsequent enhancements improved the management of higher-accuracy positioning data (Craymer 2022)
- The US National Geodetic Survey will replace NAD83 as NSRS 2022 (NATRF2022), as NAD83 has shortcomings that are best addressed through defining new datums
- This has been coming for some time, see for example Smith (2010), where "up to 2 meters of error … could cause incorrect lane determinations" (p. 7)

Canadian Spatial Reference System

Name	Type	GEOG2D	Epoch	Based.on	Implemented	US.realization	US.epoch
NAD83(CSRS96)	Static	8232	NA	ITRF94	1996-01-01		NA
NAD83(CSRS)v2	Static	8237	1997	ITRF96	1998-01-01	NAD 83(CORS96)	1997
NAD83(CSRS)v3	Static	8240	1997	ITRF97	1999-01-01	NAD 83(CORS96)	1997
NAD83(CSRS)v4	Static	8246	2002	ITRF2000	2002-01-01	NAD 83(CORS96)	2002
NAD83(CSRS)v5	Dynamic	8249	2006	ITRF2005	2006-01-01		NA
NAD83(CSRS)v6	Dynamic	8252	2010	ITRF2008	2010-01-01	NAD 83(2011)	2010
NAD83(CSRS)v7	Dynamic	8255	2010	ITRF2014	2017-05-01	NAD 83(2011)	2010
NAD83(CSRS)v8	Dynamic	10414	2010	ITRF2020	2022-11-27		NA

North American Datum

- The North American Datum realizations are not grouped as a datum ensemble, but could have been, and present the same challenges
- In successive iterations, changes known as: High Accuracy Reference Network, Federal Base Network, Continuously Operating Reference Station Network, and National Spatial Reference System have been added to NAD83, which itself replaced NAD27
- NAD83 (like WGS84) uses the GRS80 global reference ellipsoid, while NAD27 used the Clarke 1866 ellipsoid (like the *maps* and *mapproj* R packages)

North American Datum realizations EPSG remarks

```
## NAD27 [4267]: Replaced by NAD27(76) (code 4608) in Ontario, CGQ77 (code 4609) in Quebec,
##     Mexican Datum of 1993 (code 4483) in Mexico, NAD83 (code 4269) in Canada (excl.
##     Ontario & Quebec) & USA.
## NAD83 [4269]: The adjustment included connections to Greenland and Mexico but the system
##     was not adopted there. For applications with an accuracy of better than 1m replaced
##     by NAD83(HARN) in the US and PRVI and by NAD83(CSRS) in Canada.
## NAD83(HARN) [4152]: Replaced by NAD83(FBN) in CONUS, American Samoa and Guam / NMI, by
##     NAD83(NSRS2007) in Alaska, by NAD83(PA11) in Hawaii and by NAD83(HARN Corrected) in
##     PRVI.
## NAD83(FBN) [8542]: In Continental US, Puerto Rico and US Virgin Islands replaced by
##     NAD83(NSRS2007). In American Samoa and Hawaii replaced by NAD83(PA11). In Guam/NMI
##     replaced by NAD83(MA11).
## NAD83(NSRS2007) [4759]: Replaces NAD83(HARN) and NAD83(FBN). Replaced by NAD83(2011).
## NAD83(CORS96) [6783]: Replaced by NAD83(2011) (CRS code 6318) from 2011-09-06.
## NAD83(2011) [6318]: Replaces NAD83(CORS96) and NAD83(NSRS2007) (CRS codes 6783 and 4759).
```

North American Datum so far

To transform from NAD83 to NAD83(2011), lots of steps are required; this also applies to ArcGIS

```
## [1] "https://cdn.proj.org"
## Candidate coordinate operations found:  3
## Strict containment:     FALSE
## Axis order auth compl:  TRUE
## Source:  EPSG:4269
## Target:  EPSG:6318
## Best instantiable operation has accuracy: 0.1 m
## Description: NAD83 to NAD83(2011) (NADCON5, CONUS)
## Definition:  +proj=pipeline +step +proj=axisswap +order=2,1 +step +proj=unitconvert
##              +xy_in=deg +xy_out=rad +step +proj=gridshift
##              +grids=us_noaa_nadcon5_nad83_1986_nad83_harn_conus.tif +step
##              +proj=gridshift +no_z_transform
##              +grids=us_noaa_nadcon5_nad83_harn_nad83_fbn_conus.tif +step
##              +proj=gridshift +no_z_transform
##              +grids=us_noaa_nadcon5_nad83_fbn_nad83_2007_conus.tif +step
##              +proj=gridshift +no_z_transform
##              +grids=us_noaa_nadcon5_nad83_2007_nad83_2011_conus.tif +step
##              +proj=unitconvert +xy_in=rad +xy_out=deg +step +proj=axisswap
##              +order=2,1
```

North American Terrestrial Reference Frame 2022 going forward

- Since early 2025, beta versions of the modernized US NSRS have been made available, accommodating now know velocities of movements in 3 dimensions; the velocities and pre-existing inaccuracies justify modernization
- A vote on approving the modernized NSRS may occur in early to mid 2026; watch https://geodesy.noaa.gov/datums/newdatums/index.shtml
- Javier Jimenez Shaw contributed infrastructure to test the modernized NSRS in late April 2025, https://github.com/jjimenezshaw/NSRS-2022-PROJ, including a WASM option
- This makes it possible for users of US data to try to check their workflows for vulnerabilities to forthcoming changes

Testing modernized NSRS NAD83(2011) i

In the case of US data where integration of post-transition to NATRF2022 position data with pre-transition position data, it will be necessary to transform first to NAD83(2011):

```
## GEOGCRS["NAD83(2011)",
##     DATUM["NAD83 (National Spatial Reference System 2011)",
##         ELLIPSOID["GRS 1980",6378137,298.257222101,
##             LENGTHUNIT["metre",1]],
##         ANCHOREPOCH[2010]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
```

Testing modernized NSRS NAD83(2011) ii

```
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
##         AREA["Puerto Rico - onshore and offshore. United States (USA) onshore and offshore -
Alabama; Alaska; Arizona; Arkansas; California; Colorado; Connecticut; Delaware; Florida; Georgia; Idaho; Il
onshore and offshore."],
##         BBOX[14.92,167.65,74.71,-63.88]],
##     ID["EPSG",6318],
##   REMARK["Note: this CRS includes longitudes which are POSITIVE EAST. Replaces NAD83(CORS96) and NAD83(NS
```

Testing modernized NSRS NATRF2022_2D i

The NATRF2022 2D CRS definition is (for now from Javier Jimenez Shaw's test infrastructure, as yet with no assigned EPSG code) - note that it is now dynamic, but the usage block is as yet incomplete:

```
## GEOGCRS["NATRF2022",
##     DYNAMIC[
##         FRAMEEPOCH[2020]],
##     DATUM["North American Terrestrial Reference Frame 2022",
##         ELLIPSOID["GRS 1980",6378137,298.257222101,
##             LENGTHUNIT["metre",1]]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
```

Testing modernized NSRS NATRF2022_2D ii

```
##                 ANGLEUNIT["degree",0.0174532925199433]],
##             AXIS["geodetic longitude (Lon)",east,
##                 ORDER[2],
##                 ANGLEUNIT["degree",0.0174532925199433]],
##         USAGE[
##             SCOPE["Not known."],
##             AREA["World."],
##             BBOX[-90,-180,90,180]],
##         ID["NSRS","NATRF2022_2D"],
##         REMARK["NATRF2022"]]
```

Testing modernized NSRS NATRF2022 in `proj.db` i

EPSG v12.057 and earlier (including v12.029) include initial versions of "so-called" EPSG:10968 - here from v12.029:

```
## GEOGCRS["NATRF2022",
##     DATUM["North American Terrestrial Reference Frame of 2022",
##         ELLIPSOID["GRS 1980",6378137,298.257222101,
##             LENGTHUNIT["metre",1]],
##         ANCHOREPOCH[2020]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
```

Testing modernized NSRS NATRF2022 in proj.db ii

```
##              ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
##      AREA["North America - onshore and offshore: Canada - Alberta; British Columbia; Manitoba; New Brunswi
Alabama; Alaska; Arizona; Arkansas; California; Colorado; Connecticut; Delaware; Florida; Georgia; Idaho; Il
##         BBOX[23.81,167.65,86.46,-40.73]],
##     ID["EPSG",10968]]
```

from v12.049, which cautiously remarks - This data has been added to the EPSG Dataset prior to official adoption to facilitate software testing and evaluation. In the unlikely event that definitions change, the record will be deprecated and replaced with a new EPSG code:

Testing modernized NSRS NATRF2022 in `proj.db` iii

```
## GEOGCRS["NATRF2022",
##     DATUM["North American Terrestrial Reference Frame of 2022",
##         ELLIPSOID["GRS 1980",6378137,298.257222101,
##             LENGTHUNIT["metre",1]],
##         ANCHOREPOCH[2020]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
```

Testing modernized NSRS NATRF2022 in `proj.db` iv

```
##       AREA["North America - onshore and offshore: Canada - Alberta; British Columbia; Manitoba; New Brunswi
Alabama; Alaska; Arizona; Arkansas; California; Colorado; Connecticut; Delaware; Florida; Georgia; Idaho; Il
##          BBOX[23.81,167.65,86.46,-40.73]],
##     ID["EPSG",10968],
##   REMARK["CAUTION: Preliminary data from NGS beta website (https://beta.ngs.noaa.gov/). This data has bee
```

and v12.057, now without the remark, but with a new name - NATRF2022e2020, for the 2020 epoch (see also https://mastodon.social/@EvenRouault/116771634197200255) and Even Roualt's new signature: "Signed PROJ, your informal Q/A for EPSG":

Testing modernized NSRS NATRF2022 in `proj.db` v

```
## GEOGCRS["NATRF2022e2020",
##     DATUM["North American Terrestrial Reference Frame of 2022 epoch 2020.00",
##         ELLIPSOID["GRS 1980",6378137,298.257222101,
##             LENGTHUNIT["metre",1]],
##         ANCHOREPOCH[2020]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
```

Testing modernized NSRS NATRF2022 in `proj.db` vi

```
##               AREA["North America - onshore and offshore: Canada; United States (USA) - Alaska, conterminous (contiguous) lower 48 states (CONUS)."],
##           BBOX[23.81,167.65,86.46,-40.73]],
##       ID["EPSG",10968]]
```

Testing modernized NSRS - transformation pipeline i

Again, using Javier Jimenez Shaw's test infrastructure, we can access a preliminary transformation pipeline definition from NAD83(2011) to NATRF2022_2D (at the moment the axis order of the position data must be authority compliant as the test infrastructure or its use in `sf::st_transform` has not been generalized to support longitude, latitude order):

```
## Candidate coordinate operations found:  1
## Strict containment:     FALSE
## Axis order auth compl:  TRUE
## Source:  EPSG:6318
## Target:  NSRS:NATRF2022_2D
## Best instantiable operation has accuracy: 0.01 m
## Description: Conversion from NAD83(2011) (geog2D) to NAD83(2011) (geocentric) + Inverse of
##              ITRF2020 to NAD83(2011) (1) + ITRF2020 to NATRF2022 + Conversion
```

Testing modernized NSRS - transformation pipeline ii

```
##               from NATRF2022 (geocentric) to NATRF2022 (geog2D)
## Definition:  +proj=pipeline +step +proj=axisswap +order=2,1 +step +proj=unitconvert
##              +xy_in=deg +xy_out=rad +step +proj=cart +ellps=GRS80 +step +inv
##              +proj=helmert +x=1.0039 +y=-1.90961 +z=-0.54117 +rx=0.02678138
##              +ry=-0.00042027 +rz=0.01093206 +s=-5.109e-05 +dx=0.00079
##              +dy=-0.0007 +dz=-0.00124 +drx=6.667e-05 +dry=-0.00075744
##              +drz=-5.133e-05 +ds=-7.201e-05 +t_epoch=2010
##              +convention=coordinate_frame +step +proj=helmert +x=0 +y=0 +z=0
##              +rx=0 +ry=0 +rz=0 +s=0 +dx=0 +dy=0 +dz=0 +drx=4.6e-05
##              +dry=-0.000704 +drz=-4.7e-05 +ds=0 +t_epoch=2020
##              +convention=coordinate_frame +step +inv +proj=cart +ellps=GRS80
##              +step +proj=unitconvert +xy_in=rad +xy_out=deg +step
##              +proj=axisswap +order=2,1
```

Testing modernized NSRS transformation pipelines in `proj.db` i

v12.029 returns a noop for NAD83(2011) to NATRF2022:

```
## Candidate coordinate operations found:  1
## Strict containment:     FALSE
## Axis order auth compl:  TRUE
## Source:  EPSG:6318
## Target:  EPSG:10968
## Best instantiable operation has accuracy: 3 m
## Description: NAD83(2011) to WGS 84 (1) + Inverse of NATRF2022 to WGS 84 (1)
## Definition:  +proj=noop +ellps=GRS80
```

as does v12.049:

Testing modernized NSRS transformation pipelines in `proj.db` ii

```
## Candidate coordinate operations found:  1
## Strict containment:     FALSE
## Axis order auth compl:  TRUE
## Source:  EPSG:6318
## Target:  EPSG:10968
## Best instantiable operation has accuracy: 3 m
## Description: NAD83(2011) to WGS 84 (1) + Inverse of NATRF2022 to WGS 84 (1)
## Definition:  +proj=noop +ellps=GRS80
```

while v12.057 offers the noop and two similar pipelines with reported minimal loss of accuracy naming the target as NATRF2022e2020; the chosen pipeline is the same as that returned by Javier Jimenez Shaw's test infrastructure:

Testing modernized NSRS transformation pipelines in `proj.db` iii

```
## Candidate coordinate operations found:  3
## Strict containment:     FALSE
## Axis order auth compl:  TRUE
## Source:  EPSG:6318
## Target:  EPSG:10968
## Best instantiable operation has accuracy: 0 m
## Description: Conversion from NAD83(2011) (geog2D) to NAD83(2011) (geocentric) + Inverse of
##              ITRF2020 to NAD83(2011) (1) + ITRF2020 to NATRF2022e2020 (1) +
##              Conversion from NATRF2022e2020 (geocentric) to NATRF2022e2020
##              (geog2D)
## Definition:  +proj=pipeline +step +proj=axisswap +order=2,1 +step +proj=unitconvert
##              +xy_in=deg +xy_out=rad +step +proj=cart +ellps=GRS80 +step +inv
##              +proj=helmert +x=1.0039 +y=-1.90961 +z=-0.54117 +rx=0.02678138
##              +ry=-0.00042027 +rz=0.01093206 +s=-5.109e-05 +dx=0.00079
##              +dy=-0.0007 +dz=-0.00124 +drx=6.667e-05 +dry=-0.00075744
##              +drz=-5.133e-05 +ds=-7.201e-05 +t_epoch=2010
##              +convention=coordinate_frame +step +proj=helmert +x=0 +y=0 +z=0
```

Testing modernized NSRS transformation pipelines in `proj.db` iv

```
##                 +rx=0 +ry=0 +rz=0 +s=0 +dx=0 +dy=0 +dz=0 +drx=4.6e-05
##                 +dry=-0.000704 +drz=-4.7e-05 +ds=0 +t_epoch=2020
##                 +convention=coordinate_frame +step +inv +proj=cart +ellps=GRS80
##                 +step +proj=unitconvert +xy_in=rad +xy_out=deg +step
##                 +proj=axisswap +order=2,1
```

Testing modernized NSRS - Boston bounding box coordinates by CRS

Using this on the Boston census tract NAD27 data set in `spData` (Bivand et al. 2026), after first transforming to NAD83 then NAD83(2011), we can see that the coordinate values are changed, most for the big NAD27 to NAD83 step (NAD27 used the Clarke 1866 legacy ellipsoid, NAD83 GRS80 also with different datums); v12.057 gives the same transformation output as Javier Jimenez Shaw's test infrastructure:

CRS	Geodetic latitude min	Geodetic longitude min	Geodetic latitude max	Geodetic longitude max
NAD27	42.00304794	-71.52310944	42.67307281	-70.63822937
NAD83	42.0031502	-71.52261635	42.6731655	-70.63771032
NAD83(2011)	42.00315072	-71.52261689	42.67316488	-70.63771176
NATRF2022	42.00316021	-71.52262028	42.67317448	-70.63771483
NATRF2022e2020	42.00316021	-71.52262028	42.67317448	-70.63771483

Testing modernized NSRS - distances between Boston points by CRS

If we approximate using `s2` (Dunnington et al. 2026) the distances between south-west bounding box coordinates, treating them as if they had the same CRS, we can see that the modernization from NAD83(2011) to NATRF2022 causes a shift of just over a metre:

CRS	NAD27	NAD83	NAD83.2011.	NATRF2022	NATRF2022e2020
NAD27	0.00000 [m]	42.30085734 [m]	42.27365820 [m]	4.230290e+01 [m]	4.230290e+01 [m]
NAD83	42.30086 [m]	0.00000000 [m]	0.07227717 [m]	1.158747e+00 [m]	1.158747e+00 [m]
NAD83(2011)	42.27366 [m]	0.07227717 [m]	0.00000000 [m]	1.091805e+00 [m]	1.091805e+00 [m]
NATRF2022	42.30290 [m]	1.15874745 [m]	1.09180478 [m]	0.000000e+00 [m]	1.015817e-09 [m]
NATRF2022e2020	42.30290 [m]	1.15874745 [m]	1.09180478 [m]	1.015817e-09 [m]	0.000000e+00 [m]

Summary

Preliminary take-home points

- Most legacy CRS did not take continental drift into account
- The demands on CRS for accuracy and measurable errors have increased greatly, especially with the general availability of GNSS receivers
- Countries/jurisdictions adopt CRS for official purposes, and often manage them differently
- Global CRS have to match global crustal movement, but regional/national CRS are most often fixed to the most appropriate tectonic plate
- Errors at metre scale are now unacceptable, and sub-metre errors are undesirable

Concluding take-home points

- Adaptation to dynamic CRS will be required as public mapping agencies modernize their standards
- There are many public mapping agencies, and their adaptations may impact users in unexpected ways - also unexpected for the agencies
- So far, the EPSG database has not been updated carefully, so possible mis-steps can enter production inadvertently
- In addition, few tests are in place upstream to safeguard downstream users, so we need to be alert and aware that, in a dynamic world (think leap seconds - they do happen), our results are our responsibility

What is ANCHOREPOCH?

Lott (2023b), p. 57: “ANCHOREPOCH is the epoch at which a static reference frame matches a dynamic reference frame from which it has been derived. Note: Not to be confused with the frame reference epoch of dynamic geodetic and dynamic vertical reference frames. Nor with the epoch at which a reference frame is defined to be aligned with another reference frame; this information should be included in the datum anchor definition.”

Assignation in EPSG WKT2 seems somewhat confused, see the thread beginning 21 June at Time dependency in ETRFxx to ETRFyy coordinate transformations?

PR https://github.com/OSGeo/PROJ/pull/4792 adds `proj_crs_is_dynamic` to permit tests on source and destination CRS; “The terminology around all this is FUBAR, I'm afraid ” (Kristian Evers in PR)

sessionInfo i

```
> sessionInfo()

R version 4.6.0 (2026-04-24)
Platform: x86_64-pc-linux-gnu
Running under: Fedora Linux 44 (Workstation Edition)

Matrix products: default
BLAS/LAPACK: FlexiBLAS OPENBLAS-OPENMP;  LAPACK version 3.12.1

locale:
 [1] LC_CTYPE=en_GB.UTF-8       LC_NUMERIC=C               LC_TIME=en_GB.UTF-8
 [4] LC_COLLATE=en_GB.UTF-8     LC_MONETARY=en_GB.UTF-8    LC_MESSAGES=en_GB.UTF-8
 [7] LC_PAPER=en_GB.UTF-8       LC_NAME=C                  LC_ADDRESS=C
[10] LC_TELEPHONE=C             LC_MEASUREMENT=en_GB.UTF-8 LC_IDENTIFICATION=C

time zone: Europe/Oslo
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
```

sessionInfo ii

```
[1] tinytable_0.16.0

loaded via a namespace (and not attached):
 [1] s2_1.1.11               class_7.3-23            lwgeom_0.2-16           KernSmooth_2.23-26
 [5] lattice_0.22-9          digest_0.6.39           magrittr_2.0.5          evaluate_1.0.5
 [9] grid_4.6.0              fastmap_1.2.0           tmap_4.4                jsonlite_2.0.0
[13] e1071_1.7-17            leafsync_0.1.0          DBI_1.3.0               crosstalk_1.2.2
[17] cols4all_0.10           XML_3.99-0.23           codetools_0.2-20        microbenchmark_1.5.0
[21] abind_1.4-8             cli_3.6.6               rlang_1.2.0             units_1.0-1
[25] tmaptools_3.3           litedown_0.9            base64enc_0.1-6         yaml_2.3.12
[29] otel_0.2.0              leaflegend_1.2.8        tools_4.6.0             raster_3.6-32
[33] parallel_4.6.0          maptiles_0.11.0         colorspace_2.1-2        spacesXYZ_1.6-0
[37] logger_0.4.2            R6_2.6.1                png_0.1-9               proxy_0.4-29
[41] classInt_0.4-11         leaflet_2.2.3           htmlwidgets_1.6.4       terra_1.9-34
[45] data.table_1.18.4       glue_1.8.1              Rcpp_1.1.1-1.1          sf_1.1-1
[49] rnaturalearthdata_1.0.0 xfun_0.59               knitr_1.51              htmltools_0.5.9
[53] rmarkdown_2.31          leafem_0.2.5            wk_0.9.5                compiler_4.6.0
[57] sp_2.2-1                stars_0.7-2
```

Aftermatter

References i

Appel, Marius. 2026. *Gdalcubes: Earth Observation Data Cubes from Satellite Image Collections*. https://github.com/appelmar/gdalcubes.

Bivand, Roger S. 2021. "Progress in the R Ecosystem for Representing and Handling Spatial Data." *Journal of Geographical Systems* 23: 515–46. https://doi.org/10.1007/s10109-020-00336-0.

Bivand, Roger, Jakub Nowosad, and Robin Lovelace. 2026. *spData: Datasets for Spatial Analysis*. https://jakubnowosad.com/spData/.

Craymer, Michael. 2022. *NAD83(CSRS): From Static to Dynamic*. Canadian Geodetic Reference System Committee. https://cgrsc.ca/wp-content/uploads/2022/04/NAD83CSRS_AOLS_20220302_EN.pdf.

Dunnington, Dewey, Edzer Pebesma, and Ege Rubak. 2026. *S2: Spherical Geometry Operators Using the S2 Geometry Library*. https://r-spatial.github.io/s2/.

Evenden, Gerald I., Even Rouault, Frank Warmerdam, et al. 2026. *PROJ*. V. 9.8.1. Zenodo, released April. https://doi.org/10.5281/zenodo.19497389.

Evers, Kristian, and Thomas Knudsen. 2017. *Transformation Pipelines for PROJ.4*. https://www.fig.net/resources/proceedings/fig_proceedings/fig2017/papers/iss6b/ISS6B_evers_knudsen_9156.pdf.

Häkli, Pasi, Kristian Evers, Lotti Jivall, et al. 2023. *Journal of Geodetic Science* 13 (1): 20220155. https://doi.org/10.1515/jogs-2022-0155.

Hijmans, Robert J., Andrew Brown, and Márcia Barbosa. 2026. *Terra: Spatial Data Analysis*. https://rspatial.org/.

Iliffe, Jonathan, and Roger Lott. 2008. *Datums and Map Projections: For Remote Sensing, GIS and Surveying*. CRC.

Kainu, Markus, Joona Lehtomaki, and Leo Lahti. 2026. *Geofi: Access Finnish Geospatial Data*. https://ropengov.github.io/geofi/.

Knudsen, Thomas, and Kristian Evers. 2017. *Transformation Pipelines for PROJ.4*. https://meetingorganizer.copernicus.org/EGU2017/EGU2017-8050.pdf.

Konyk, William, Alex Smith, Bob Wong, and Andrew Tollefson. 2025. "Thirty Years of Maintaining WGS 84 with GPS." *NAVIGATION: Journal of the Institute of Navigation* 72 (2). https://doi.org/10.33012/navi.693.

Lott, Roger. 2015. *Geographic Information—Well-Known Text Representation of Coordinate Reference Systems*. Open Geospatial Consortium. http://docs.opengeospatial.org/is/12-063r5/12-063r5.html.

Lott, Roger. 2019. *Geographic Information—Well-Known Text Representation of Coordinate Reference Systems*. Open Geospatial Consortium. https://docs.ogc.org/is/18-010r7/18-010r7.html.

Lott, Roger. 2023a. *Geographic Information—Well-Known Text Representation of Coordinate Reference Systems*. Open Geospatial Consortium. https://docs.ogc.org/is/18-010r11/18-010r11/pdf.

Lott, Roger. 2023b. *OGC Abstract Specification. Topic 2: Referencing by Coordinates*. Open Geospatial Consortium. https://docs.ogc.org/as/18-005r8/18-005r8.pdf.

Pebesma, Edzer. 2026a. *Sf: Simple Features for r*. https://r-spatial.github.io/sf/.

Pebesma, Edzer. 2026b. *Stars: Spatiotemporal Arrays, Raster and Vector Data Cubes*. https://r-spatial.github.io/stars/.

Pebesma, Edzer, and Roger Bivand. 2023. *Spatial Data Science with Applications in R*. CRC Press. https://r-spatial.org/book.

Pebesma, Edzer, and Roger Bivand. 2026. *Sp: Classes and Methods for Spatial Data*. https://github.com/edzer/sp/.

Rouault, Even, Frank Warmerdam, Kurt Schwehr, et al. 2026. *GDAL*. V. v3.13.0. Zenodo, released May. https://doi.org/10.5281/zenodo.20083023.

Smith, Dru. 2010. *Improving the National Spatial Reference System*. National Geodetic Survey. https://www.ngs.noaa.gov/2010Summit/Improving_the_NSRS.pdf.

Sumner, Michael. 2026. *Vapour: Access to the Geospatial Data Abstraction Library (GDAL)*. https://github.com/hypertidy/vapour.

Sumner, Michael, and Charles Karney. 2026. *Geographiclib: Access to GeographicLib*. https://github.com/hypertidy/geographiclib.

Tennekes, Martijn. 2026. *Tmap: Thematic Maps*. https://github.com/r-tmap/tmap.

Toney, Chris, and Michael D. Sumner. 2026. *Gdalraster: Bindings to GDAL*. https://firelab.github.io/gdalraster/.